

Dance

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Introduction

Dancers are easily spotted on the street, as their bodies seem to show a special “connectivity” that allows for harmonious, continuous and elegant movement. There are many different factors contributing to this impression, one of which might be the state of the fascial system. Dance in its variety of different styles offers a great source of body and movement expertise, gained through physical experience. By teaching it, the expertise of generations of dancers and dance teachers is passed on in the studios being adapted anew by each individual dancer and thus practically tested over a wide range of different body types. However, big sections of this knowledge have not yet been proven scientifically.

Since the 1980s, there has been an increasing interest in dance from a medical perspective. In its early years, the new field of dance medicine focused mainly on the dancer’s health, his/her working conditions, typical dance injuries and injury prevention (Laws et al., 2005; Hincapié et al., 2008; Simmel, 2014). Analogous to the development in sports medicine, where the focus has changed from the management of sports injuries to a broader definition of a medicine of exercise, dance medicine is widening its horizons towards the dancer’s body expertise, motor learning abilities and specific training methods (Wyon et al., 2016). It is the expanded knowledge from medical research, sports medicine and exercise science that allows for a deeper scientific understanding of many traditional dance training methods.

Dancers train to combine a high degree of flexibility, exceptional balance (Bläsing et al., 2018) and remarkable coordination of whole-body movements (Bronner, 2012) with suppleness and elegance. The flow of multidimensional movements and the constant variation of exercise patterns may play an important role in stimulating the fascial system.

Dancers often describe their dance training as an outstanding training for body connectivity. As a dancer in transition to her second profession states: “Searching in all varieties of sports, I could not find any other athletic training which provides the same feeling of a whole-body training as dance training does. After a ballet class my body feels trained from head to toe”. The ongoing research on fascial tissue might help to find a scientific explanation for this body experience. Meanwhile, looking at dance training from a fascial training perspective can expand today’s training recommendations by highlighting ideas which might increase the effectiveness of a fascial-orientated fitness training.

Dance training

In contrast to many sports, where the attention is focused on achieving measurable external goals, dance emphasizes not only the outer shape and the outcome of the movement but also the perception and awareness of the sensations within the dancer’s body (De Lima, 2013). It is the ability to perceive, classify and react to one’s individual inner body perspective that creates a

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sophisticated dancer. Dance seems to be a special combination between strong athletic performance ability and high body perception and awareness.

When speaking of dance and dance training, one has to keep in mind the enormous variety of different dance techniques. From classical ballet, modern and contemporary dance to urban, break dance or tap, many different dance styles are shown on stage and might be part of a choreography. This great diversity in dance styles places high demands on the dancer's body. Obviously, there is not one specific training method that can provide everything dancers need for their fitness (Angioi et al., 2009). However, even today, for many dancers the classical ballet technique seems to be the basis of their dance career as well as their daily training routine. Even dance companies who mainly perform contemporary dance or dance theatre tend to offer daily ballet classes to their dancers. Enriched with somatic principles and contemporary dance elements, these classes seem to offer helpful training methods, most of them being developed practically through trial and error, yet still do not have scientific proof.

"Somatic movement education" (Hanna, 2004) describes the process of a sensory motor training creating improved muscle function, enhanced sensory awareness and adaptation of muscle patterns. With its integration, listening to the body and responding to perceived sensations by consciously questioning movement habits and altering movement patterns became an important tool in teaching and learning dance. Dance and somatic approach influenced each other mutually with the dancer's experience supporting somatic investigation and the somatic approach being applied to the dance technique (Bartenieff and Lewis, 1980). By addressing the inner body perspective, dance stimulates both

the proprioception and interoception of the body. The highest density of proprioceptive and interoceptive receptors are found within the fascial tissue (Schleip et al., 2012B), so it seems to play an important role in the perception of inner sensations. It can, hence, be assumed that training body awareness is deeply associated with the fascial system.

Adaptation of fascial tissue

"Form follows function". This statement by the American sculptor, Horatio Greenough, which addressed the organic principles in architecture, is quoted in many bodywork methodologies to refer to the impressive adaptability of the body. How the body is used and the way the individual tissues are put under strain influences its formation and architecture. It is this principle that, for example, explains the tightening of the iliotibial band at the outside of the thigh in many classical trained dancers as the turnout position of the leg increases the tension on this structure.

Research has shown that fascial tissue is particularly adaptable to regular strain. When put under increasing physiological stress, it reacts to the loading patterns by remodeling the architecture of its collagenous fiber network, resulting in a change in length, strength, elasticity, and an increasing ability to withstand shearing forces (Schleip and Müller, 2013). By contrast, aging processes and lack of movement—at least in animals—lead to a more haphazard and disoriented arrangement of the collagenous fibers, diminishing the elasticity of the fascial tissue (Järvinen et al., 2002). Thus the local architecture of the fasciae reflects the individual history of previous strain and movement demands.

To achieve adaptation in human fasciae, applied strains need to exceed the degree of normal daily life activities. This requires a specific training that stimulates the fibroblasts to build

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up an elastic fiber architecture. Movements that load the fascial tissues over diverse whole-body extensions using their elastic springiness for fast form and position changes seem to be especially efficient. There are many examples in dance, like high jumps starting from a deep squat position or leg extensions that proceed in contra directional upper body and arm movement with fast direction changes. As many dance exercises require a great range of motion and a variety of movement angles, dance offers a large repertoire of exercises potentially helping to maintain and train the strength, elasticity and shearing ability of the fascial tissue.

Dance offers both training and challenge to the fascial network. Dance sequences are, in general, practiced on both sides, right and left, as well as forwards and backwards. By using these multidirectional whole-body movements, dance trains a wide variety of motions, which calls for a high shearing ability of the different fascial layers. Additionally, by putting specific focus on alignment and placement, dance supports to counteract potential harmful asymmetries within the body, for example, like a hyperlordosis of the lumbar spine, which could lead to a tightening of the thoracolumbar fascia. With targeted flexibility training being an integral part of most dance classes, dance seems to keep the soft tissue compliant and extensible. Counter-movements, dynamic muscle loading and springy motions are part of many dance steps, challenging the elastic storage capacity of the fascial tissue. Furthermore, by focusing on inner body awareness while performing exercises, dance stimulates proprioception.

In summary, dance engages the fascial system through:

- Multidirectional whole-body movements
- Specific focus on alignment and placement to counteract body asymmetries

- Targeted flexibility training
- Counter-movements, dynamic muscle loading and springy motions
- proprioceptive stimulation.

Dance as fascial training

Although results from fascia research are only beginning to be included specifically, many dance elements, exercises and corrections address the fascial connective tissue and correspond with the principles of fascial fitness training (Schleip and Müller, 2013).

Aspects of dance as fascial training:

- Myofascial release
- Stretching
- Preloading – counter-movements
- Cyclic training
- Body awareness.

Myofascial release

Findings from animal studies suggest that immobility quickly fosters the development of additional cross-links, which render the fascial tissue less flexible (Järvinen et al., 2002). It seems a natural conclusion that whole-body mobility is an important prerequisite for the elasticity and shearing ability of the fascial system. In our daily movement habits, we are a long way away from using the full range of motion our joints would permit. Instead, we tend to stick to familiar movement patterns which, over time, might limit our general range of motion and reduce mobility. This is where the targeted mobilization exercises of dance come to the fore.

Exercise: Release of the dorsal fascia

This exercise is thought to gradually release the dorsal fasciae, starting from the cervical area all

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the way down over the heels to the plantar side of the toes. Start by standing upright, feet parallel and hip-width apart, with the hands crossing at the back of the head. Slowly roll down, vertebra by vertebra, flexing the spine, starting from the top of the head. When an initial stretch can be felt at the back of the spine, stop and gently increase the pressure on the head, resisting



Figure 41.1

Release of the dorsal fasciae

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by performing an isometric contraction of the back muscles. Hold the contraction for 8 s, then release the tension and slowly continue rolling down, stopping at any newly felt stretch. Slowly proceed in this manner until reaching the deepest possible stretching position (Figure 41.1).

Exercise: Release of the plantar fascia

This exercise focuses on releasing the plantar fascia of the feet while preloading the dorsal fasciae. Start by standing in the yoga downward-facing-dog position, feet parallel and hip-width apart. Feel the connection between the soles of the feet and the hands flat on the floor, focusing on the stretch of the dorsal fasciae by lifting up the sit bones high in the air. Shift the weight onto the hands and slowly walk step-by-step towards the hands using the mobility of the foot and the elasticity of the sole to deepen the stretch on the dorsal fascia (Figure 41.2).

Stretching

The collagen fibers of the fascial system are primarily shaped by tensional strain rather than by



Figure 41.2

Release of the plantar fascia

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compression (Schleip and Müller, 2013). This fact, and the decrease in range of motion that usually accompanies aging, reducing the shearing ability between distinct fascial layers, leads to a strong recommendation for regular stretching in order to remodel and train the fascial tissue. Using variations of different stretching styles seems to be more effective than sticking to only one method. Thus, alternating between slow passive stretches at different angles and dynamic and bouncing stretches has been recommended. Research suggests that fast dynamic stretching is even more effective as a fascial training when combined with a preparatory counter-movement (Fukashiro et al., 2006).

Dance uses a wide variety of stretching and flexibility methods throughout its training. Passive stretching methods are common to increase general flexibility. The typical multidirectional movements, with slight changes in angle while deepening the stretch, are beneficial for the fascial system. Active and dynamic stretching are naturally integrated into most dance steps. Each controlled high extension of the leg and each leg-kick elongates the muscular and fascial tissue, placing tensional strain on the collagenous fibers. The dynamic swinging used in leg swings, whether lying on the floor or standing, stimulates many epimysial tissues in the leg and pelvis, which are extended beyond their habitual conditions. In dance, all these stretches engage myofascial chains extending over multiple joints.

Exercise: Dynamic stretching of the back of the leg

This exercise uses a dynamic muscular loading pattern in which the muscle is briefly activated while in its lengthened position. For this purpose, Schleip and Müller (2013) propose soft elastic bounces at the end ranges of available motion. Start by standing upright, feet parallel

and hip-width apart, with the balls of the feet standing on a stair. Be aware of the connection between heel and head letting the head easily “float” on the cervical spine, and slowly lower the heel. At the end of the available movement, perform soft elastic bounces, lowering the heel even further. Finish the exercise with a short isometric contraction of the calf muscles while still in the deepest stretching position (Figure 41.3).

Preloading – counter-movements

The elasticity of the fascial tissue is key to its high capacity to store kinetic energy. Using the dynamic catapult effect of the collagen fibers allows for an energetic and elastic movement, not only when jumping and running, but also in daily activities, i.e. walking. The higher the demands on velocity and momentum, the more the elasticity



Figure 41.3

Dynamic stretching of the back of the leg

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of the fascial system comes into account. In a healthy fascial system, actively preloading the fascial tissue before an action facilitates the dynamic impulse. This can be used by starting a movement with a slight counter-movement in the opposite direction. While performing the actual movement, the stored energy will dynamically release in a passive recoil action, thereby using the catapult effect of the connective tissue.

Preparatory counter-movements are common in dance. Many dance exercises implement the energy-loading momentum as preparation for a dance step. Using a deep demi-plié (bending the knees without lifting the heels off the floor) before jumping or in preparation for a pirouette, loads the soft tissue and allows for a dynamic passive recoil to start the actual movement. The principles of fall, rebound and recovery, as used in many contemporary dance techniques (Diehl and Lampert, 2011), seem an ideal training for the preloading and recoil capacity of the fascial tissue. Breathing in before bending backwards increases the elastic tension in the upper body and thus elongates the spine, diminishing the pressure on the discs as well as on the intervertebral joints.

Exercise: Leg swings

The focus of this exercise is on the preloading of the posterior fascial chain. Start by lying supine, both knees bent, feet parallel and aligned with the sit bones. The right knee drops to the left side. The momentum of the movement allows the lower right leg to swing along the floor to the left, thereby also rotating the pelvis to the left. The entire right posterior fascial chain from the shoulder to the foot becomes preloaded, supporting the right leg, at the end of its loading capacity, to swing back along the floor to the starting position. The left side takes over the momentum, starting to swing to the right, preloading the left posterior fascial chain. Repetitive alternation

between right and left swing preloads the fascial chains and facilitates the passive recoil action at the end of the preloading capacity (Figure 41.4).

Cyclic training

Experimental research shows that fluid is pressed out of the fascial tissue when subjected to strain, similar to a sponge being squeezed. When releasing the strain, this area will refill with fluid, which comes from the surrounding tissue, the lymphatic network and the vascular system (Schleip et al., 2012A). Although the optimal durations for the loading and unloading phases have yet to be elucidated, their proper timing is important to facilitate optimal rehydration, as the short pause gives the tissues the chance to absorb nourishing fluid, thereby boosting the fascial metabolism. A cyclic training character that alternates periods of intense strain with targeted breaks is therefore recommended.



Figure 41.4

Leg swings

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In general, rhythm plays a vital role in dance: be it using music as an external “pacemaker”, focusing on one’s individual breath pattern or engaging with the “inner rhythm” of the movement itself. Rhythmically loading and unloading the fascial tissue comes naturally in dance, alternating between stretching and releasing, between elongating and contracting. Thus many dance exercises implement a cyclic fascial training and hence support fascial health.

Body awareness

Good coordination and awareness of one’s whole-body connectivity are essential keystones for dance. Allowing movements to fluently continue through the body, focusing on maintaining the alignment and expanding the body in space, brings attention to the fascial network. Dance, especially contemporary dance, implements a large repertoire of fascial awareness exercises, which focus on the body’s inner communication network and the stretching of the connecting tissue.

Exercise: Rocking the dorsal fasciae on the floor

Focus on the dorsal fasciae to facilitate their mobility. Start by lying supine, both legs parallel with the feet in dorsal flexion, anchoring the heels into the floor. Rhythmically point and flex the feet allowing the momentum to continue up to the head, gently flexing and extending in the most upper cervical joint. Feel the connectivity from the toes up to the head and register the impulses of the feet as they induce the motion of the head.

Exercise: Awareness of whole-body connectivity

Elongate and mobilize the lateral myofascial chain: lying supine with legs stretched, and arms opened out to the side with palms facing towards the ceiling, start with the impulse of

the right foot slowly pulling the right leg across to the left side, allowing the pelvis and spine to follow. Feel the resistance by holding the right arm and shoulder on the floor. To perform the stretch dynamically, alternate the pulling impulse between right leg and right arm. Roll back to the starting position, taking time to compare both sides of the body, then continue the exercise on the left side (Figure 41.5).

Supplementary fascial training through dance

In addition to offering a large selection of exercises, dance also provides helpful supplementary ideas to support fascial training. Although these practices have been inspired by physical feedback and are not yet proven scientifically, they have been applied practically by generations of dancers and have helped dancers to gain their specific “whole body connectivity” for which they are renowned. By performing most of the exercises on both sides, dance provides an “equilibrium” for the body. Regarding fascial training, there are four movement



Figure 41.5

Awareness of whole-body connectivity

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principles, which are fundamental to almost every dance training and which, in application to the specific fascial training, could increase its effectiveness.

Increase fascia-oriented training training by the use of dance-specific movement principles:

- Awareness of oppositions
- Elongation
- Counter-movements
- Momentum and rebound.

The awareness of opposition is key for body connectivity. As a preparation for the exercise, the dancer is focusing on the specific body parts affected by the movement to follow, their connection and their separation. This allows for the feeling of space and elongation, yet connection between the opponents. For example, when bending the upper body to the right side, the left foot firmly keeps its contact to the floor. The two opponents, head to the right side and left foot, stay connected, thus allowing elongation and stretching of the fascial tissue on the left side of the body.

When lifting one leg off the floor or using the arms for expressive dance movements, dancers tend to focus on the elongation of their extremities. With the image of widening the movement into space, filling out the surrounding area, the dancer increases their bodily borders. From the perspective of fascial training, this elongation of the extremities allows for further loading and prestretching of the fascial tissue.

Keeping one's balance is crucial in dance. To achieve a natural equilibrium, regardless of the performed dance movement, the dancer makes use of small counter-movements that allow for

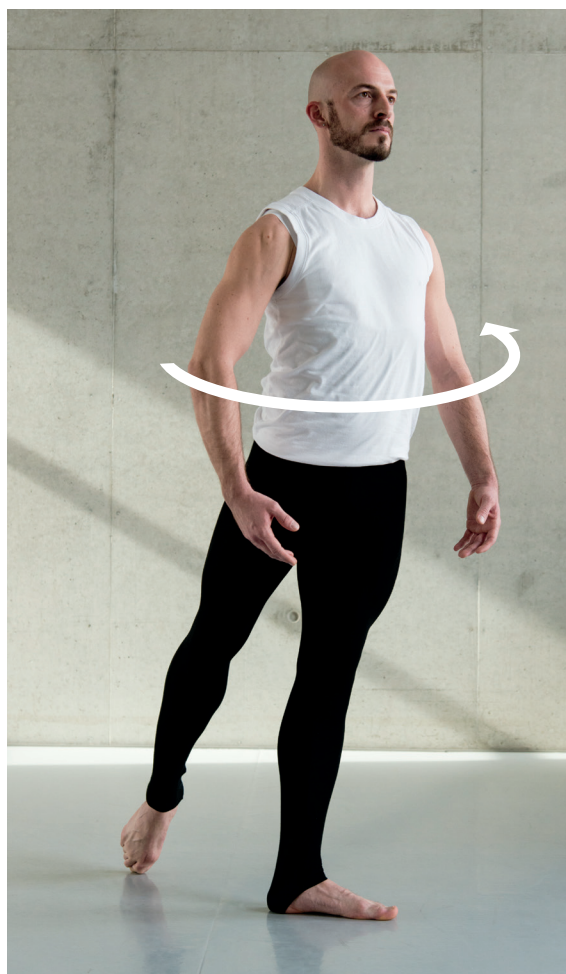


Figure 41.6

In order to achieve an economic equilibrium, the dancer performs a small rotation of the spine to the left when lifting the right leg to the back.

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stability and counter-strain. For example, to achieve economic equilibrium, a small rotation of the spine to the left may be used when lifting the right leg to the back (Figure 41.6). Fascially speaking, this small counter-movement allows loading of the corresponding fascial tissue.

Jumping, rolling on the floor, or lifting, each of these dance movements demands perfect timing of the impulse, momentum and rebound: the optimal timing in releasing the preloaded muscular and connective tissues. A cat-like jump with a noiseless landing requires a high level of elasticity within the fascial tissue, using the body like a spring, preloading its fibers in preparation for the movement. Applying this momentum and rebound to daily life activities, for example, by using the elastic rebound while walking fast or jumping up stairs, allow for an implemented fascial fitness training.

Although these dance movement principles still require further research regarding their

effect on the fascial tissue, applying them into fascial training might add helpful aspects and increase its effectiveness.

Dance uses a wide variety of fascial training techniques and principles, which are nowadays recommended and supported by an increasing body of evidence. In its practical application, dance provides ideas for further research in fascial fitness training. However, with its highly complex movement patterns and its whole-body integration, dance remains challenging for scientific research. Yet further investigation on the dancer's body expertise might help to deepen the understanding of fascial structures and their trainability.

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